J&K; EduSetu

Your Bridge to Education & Opportunities

SIH 2026 Presentation Strategy & Complete Platform Guide

Problem Statement	SIH25094 — Government of Jammu & Kashmir
Theme	Smart Education — Student Empowerment & Career Guidance
Team	Team Error404 — NIE Mysuru · CSE · Batch 2027
Purpose	Complete app details, 6-slide PPT blueprint with timing, 5-minute script, live demo strategy, and judge Q&A; preparation

1. The Problem Statement & Ground Reality

Jammu & Kashmir has over **2 million students** in secondary and higher education, yet fewer than **200 certified career counselors** across the entire UT. This creates a devastating information asymmetry where students in remote mountainous districts have virtually zero access to career guidance, scholarship information, or college admission data.

The Five Critical Bottlenecks

• **Connectivity Crisis:** Mountainous terrain limits most rural J&K; to 2G/3G speeds (0.1–0.5 Mbps). Any solution must work offline or at near-zero bandwidth.

• **Scheme Complexity:** Central & state schemes (PMSSS, NSP, JKCET, INSPIRE) have layered eligibility criteria that students and parents cannot navigate alone.

• **LLM Hallucinations:** Generic AI chatbots fabricate cutoff marks, invent scholarship deadlines, and hallucinate college data — dangerous for life-altering decisions.

• **Zero Interview Prep:** No mock interview infrastructure in remote districts. Students face JKSSB, KAS, and placement interviews with zero practice.

• **High Dropout Risk:** 15%+ dropout rates with no early warning system. Institutions lack tools to identify at-risk students before it's too late.

2. What is J&K; EduSetu?

J&K; EduSetu (*'Setu'* = Bridge) is an **AI-powered, offline-capable, zero-cost regional education intelligence platform** purpose-built for J&K; students. It operates across three modes:

Mode	How It Works	Latency
■ 2G Ultra-Lite	Pre-computed government data, zero API calls, keyword matching	< 10ms
■ Smart Auto-Detect	Checks 2G cache first, routes to AI cloud only when needed	10ms – 3s
■ AI Cloud	Full RAG pipeline: ChromaDB retrieval → Gemini/Groq generation	2–5s

Unlike generic chatbots, every EduSetu answer is **grounded in verified government documents** with inline citations showing source name, page number, and similarity score.

3. Complete Feature Inventory (11 Modules)

#	Module	File	Key Capabilities
1	RAG Career Advisor	rag_engine.py	ChromaDB vector search, 494+ chunks, Gemini/Groq generation, cited answers, follow-up detection, caching
2	2G Offline Engine	offline_engine.py	22 pre-computed topics, sub-10ms responses, keyword matching, portal deep-links, zero API calls
3	Smart Model Router	model_router.py	3-tier complexity classification (simple/medium/complex), session-level quota tracking
4	Error Diagnostics	error_handler.py	API key / rate limit / timeout / ChromaDB error cards with friendly recovery actions
5	College Explorer	college_data.py	22+ J&K; institutions, 10 districts, seat matrices (OM/SC/ST/RBA), geographic map, cutoff comparison

6	Scholarship Engine	scholarship_engine.py	15 schemes, 7-field eligibility matching, deadline calendar, document checklists
7	Job Intelligence	job_intelligence.py	30+ job profiles (JKSSB/JKPSC/JK Bank/Central/Private), 50+ skill-career mappings, exam prep roadmaps
8	Mock Interview	mock_interview.py	5 templates (JKSSB, KAS, Engineering, Medical, Bank PO), 4-dimension AI rubric scoring
9	Resume Analyzer	resume_analyzer.py	PDF extraction, 9 section types, 4-dimension scoring (100 pts), AI narrative review, job-fit comparison
10	Student Analytics	student_analytics.py	8-factor dropout risk scoring, 4-tier classification, automated intervention plans, cohort dashboard
11	Streamlit UI	app.py	1500+ lines, 7 navigation tabs, trilingual (EN/HI/UR), glassmorphism CSS, mobile-responsive, interactive J&K; map

Knowledge Base Coverage

Category	Count	Highlights
Scholarships	15 schemes	PMSSS, Post-Matric NSP, INSPIRE, Central Sector, Minority MCM, AICTE Pragati/Saksham, Maulana Azad, GATE, UGC NET JRF, Vidyalakshmi, NMDFC
Colleges	22+ institutions	NIT Srinagar, IUST, SSM, GMC (7 campuses), SKIMS (2), GAMC, GUMC, KU, JU, CUK, CUJ, SKUAST-K/J, GDC Law
Jobs	30+ profiles	JKSSB (8), JKPSC (5), JK Bank (3), Central (6), Private (5), Entrepreneurship (2)
Documents	17 files / 494 chunks	Official government PDFs and curated text files indexed in ChromaDB

4. Technical Architecture & Innovation Stack

Technology Stack

Layer	Technology	Purpose
Frontend	Streamlit 1.42+	Glassmorphism UI, Plus Jakarta Sans, 7-tab navigation, mobile-responsive
Primary LLM	Google Gemini (3 tiers)	Smart complexity routing: Flash Lite → Flash → Flash Pro
Fallback LLM	Groq (Llama 3 8B)	Zero-downtime cloud fallback when Gemini hits rate limits
Vector Store	ChromaDB (Persistent)	494+ chunks, HNSW cosine similarity, MD5 manifest fingerprinting
Embeddings	all-MiniLM-L6-v2	Local semantic search, singleton memory cache
PDF Parsing	pypdf	Government document ingestion + resume text extraction
Tokenization	tiktoken (cl100k_base)	300-token chunks with 50-token overlap
Framework	LangChain	LLM chain orchestration for Gemini and Groq

Key Technical Innovations (Judge Differentiators)

- **99.87% Startup Speedup:** MD5 manifest fingerprinting skips redundant ChromaDB re-indexing. Launch reduced from ~28s to 35ms.
- **Zero-Hallucination Architecture:** Every LLM response is grounded against retrieved ChromaDB chunks with source citations, page numbers, and similarity scores.
- **Dual-Path Resilient Execution:** Gemini → Groq → ChromaDB direct → 2G cache. Four-layer fallback. Student NEVER sees a crash.
- **Follow-up Intelligence:** Detects reformatting queries ('make it shorter', 'elaborate') and bypasses redundant ChromaDB retrieval.
- **Trilingual Adaptive Prompting:** Language toggle (EN/HI/UR) injects system-level instructions so AI responses match the student's language.
- **8-Factor Dropout Risk Engine:** Weighted composite scoring across attendance, grades, income, first-gen status, remote district, and engagement. 4-tier classification with automated intervention plans.

5. Aisha's Story — For the Impact & Benefits Slide

Aisha's story is a **single-slide case study** used in Slide 5 to illustrate real-world impact. It is NOT the throughline of the presentation — it is a focused emotional beat that grounds the technical demo in a human outcome.

The Before (Left Column of Slide 5)

Aisha Bano, 18, Kupwara District, Kashmir Valley
Scored 89% in Class 12 PCM. Father earns ■1.8L/year. First in family to finish school.

■ School has **zero career counselors**. Nearest guidance center is 120 km away in Srinagar.
■ Phone runs on **2G**. Google searches time out after 45 seconds.
■ Asked ChatGPT about PMSSS — told her the deadline had **already passed**. (It hadn't.)
■ Has **no idea** she qualifies for 5 different scholarships worth ■6+ lakh.
■ Has **never practiced** for an interview or had her resume reviewed.
■ Father wants her to **drop out and work** because he doesn't know financial aid exists.

The After (Right Column of Slide 5)

After opening J&K; EduSetu on her 2G phone:

■ Types 'PMSSS scholarship' → verified answer in **0.2ms**. No loading, no timeout. Direct portal link.
■ Fills 7-field profile → matched to **5 scholarships worth ■6.2 lakh**: PMSSS, INSPIRE, Central Sector, Post-Matric, AICTE Pragati.
■ Explores NIT Srinagar's CSE seat matrix — sees 38 OM seats, JEE rank ~22,000 required. Discovers IUST as a backup.
■ Practices **AI mock interview** — scored on Accuracy, Clarity, Completeness, Confidence. Runs 3 sessions.
■ Uploads resume → scored 52/100 → AI says 'add projects and certifications' → revises to 71/100.
■ Father sees the ■1L/year PMSSS confirmation. **Supports her B.Tech decision.**

Bottom-line callout for the slide: *"■6.2 lakh in financial aid discovered in under 10 minutes on a 2G phone."*

Delivery note: Spend ~20 seconds on 'Before', ~25 seconds on 'After'. Let the contrast do the emotional work. Don't over-narrate.

6. The 7-Slide PPT Blueprint (Including References)

You have **5 minutes** and **7 slides** (6 content + 1 references). Each content slide should have max 5–6 bullet points and one visual. The PPT is your visual aid — YOU are the presenter. Don't read slides.

SLIDE 1 — PROBLEM STATEMENT (45 seconds)

Title: "2 Million Students, Zero Counselors"

• Visual: Infographic showing J&K; stats — 2M students vs <200 counselors (10,000:1 ratio)

• 5 bottleneck icons: ■ 2G connectivity | ■ Complex schemes | ■ AI hallucinations | ■ No interview prep | ■ 15% dropout

• Key stat card: "A student in Kupwara has a 1-in-10,000 chance of accessing a career counselor."

• Closing question: "What if we could give every student an AI counselor that works on 2G?"

DELIVERY: Open strong with the 2M vs 200 stat. Let the disparity sink in. Pose the closing question directly to judges.

DO NOT open with team intro or problem statement ID. Hook first, credentials on Slide 2.

SLIDE 2 — OUR SOLUTION — J&K; EduSetu (45 seconds)

Title: "J&K; EduSetu — Your Bridge to Education & Opportunities"

• Visual: Clean app screenshot showing the 7-tab navigation interface

• 3 differentiator pillars: (1) Works on 2G in <10ms (2) 100% cited government data (3) Free, forever

• Module icon row: ■ AI Advisor | ■ Colleges | ■ Scholarships | ■ Jobs | ■ Interview | ■ Resume | ■ Admin

• Footer badge: SIH25094 · Team Error404 · NIE Mysuru · CSE · Batch 2027

DELIVERY: 'We built EduSetu — Setu means bridge. Three things make it different from anything out there.' Then list the 3 pillars.

SLIDE 3 — LIVE DEMO (90 seconds — LONGEST SLIDE)

THIS IS NOT A CONTENT SLIDE — this is your LIVE APP DEMO on localhost.**

• Demo flow (rehearse this 10+ times):

• → 2G Mode: Type 'PMSSS scholarship' → instant 0.2ms answer with portal link (15s)

• → Scholarship Engine: Fill profile (PCM, ■1.8L, OM, Female, 89%) → 5 matches appear (20s)

• → College Map: Click ■ tab → interactive J&K; map → 26 pins across Valley & Jammu (10s)

• → Visual Seat Bar: Scroll to NIT Srinagar → colored OM/SC/ST/RBA distribution (10s)

• → Trilingual Toggle: Switch to ■■■■ → UI updates to Urdu (10s)

• → Mock Interview: Start JKSSB → submit one answer → AI score with rubric breakdown (25s)

CRITICAL: Have a backup 90-second screen recording. If WiFi/projector fails, play the video.

NEVER say 'let me find it' or 'wait, where is that'. Every click must be pre-rehearsed.

SLIDE 4 — TECHNICAL ARCHITECTURE (45 seconds)

Title: "Under the Hood — Dual-Path RAG & Edge Architecture"

• Visual: Architecture diagram showing the dual path (2G Offline ↔ Cloud RAG ↔ Gemini/Groq fallback chain)

• Top 3 innovation callouts:

• 1. 99.87% startup speedup via MD5 manifest fingerprinting (28s → 35ms)

• 2. Zero-hallucination RAG with inline source citations (document + page + similarity score)

• 3. 3-tier model complexity routing — routes cheap queries to cheapest Gemini tier

• Tech badge row: Gemini · ChromaDB · LangChain · Streamlit · pypdf · tiktoken · all-MiniLM-L6-v2

• Test badge: 19/19 automated tests passing

DELIVERY: 'Judges want to know you BUILT this.' Emphasize manifest fingerprinting and the 4-layer fallback chain.

SLIDE 5 — IMPACT & BENEFITS (Aisha's Before/After) (45 seconds)

Title: "From Information Desert to Empowerment"

• Visual: Split-screen Before/After layout with Aisha's case study

• LEFT (Before — Red/Orange tint): No counselor ■ | 2G timeout ■ | ChatGPT hallucination ■ | Zero interview prep ■ | Father says drop out ■

• RIGHT (After — Green tint): 5 scholarships (■6.2L) ■ | 0.2ms instant answer ■ | Cited govt data ■ | AI-scored mock interview ■ | Father supports B.Tech ■

• Bottom callout: "■6.2 lakh in financial aid discovered in under 10 minutes on a 2G phone."

• Quantified metrics row: 15 scholarships | 22+ colleges | 30+ careers | 3 languages | <10ms offline | 19/19 tests

DELIVERY: ~20s on Before, ~25s on After. Don't over-narrate — let the contrast speak. The ■6.2L number is your mic-drop stat.

SLIDE 6 — SCALABILITY & CALL TO ACTION (30 seconds)

Title: "From J&K; to Every Underserved State"

• Visual: India outline map with J&K; highlighted (current) → arrows to Rajasthan, Chhattisgarh, NE India (future)

• Key insight: "EduSetu's architecture is state-agnostic. Replace the knowledge base — the engine generalizes."

• Scalability path: Phase 1 (J&K;) → Phase 2 (3 more states, 6 months) → Phase 3 (B2G licensing, 12 months)

• Closing line: "Two million students need a bridge. We built it. Thank you."

• Footer: Team Error404 | NIE Mysuru | SIH25094 | GitHub link

DELIVERY: End clean and confident. 'Two million students need a bridge. We built it. Thank you.' PAUSE. Eye contact. Done.

SLIDE 7 — RESEARCH & REFERENCES (SIH Format) (Displayed, not presented verbally)

Title: "References & Research Foundation"

Format: IEEE citation style — numbered list, authors, title, journal/conference, year.

Layout: Two columns — Research Papers (left), Government Sources & Technologies (right).

• Research Papers / Academic References:

• [1] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS, 2020.

• [2] J. Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers," NAACL-HLT, 2019.

• [3] W. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," EMNLP, 2019.

• [4] T. Brown et al., "Language Models are Few-Shot Learners (GPT-3)," NeurIPS, 2020.

• [5] Gemini Team, Google, "Gemini: A Family of Highly Capable Multimodal Models," arXiv:2312.11805, 2023.

• Government / Institutional Data Sources:

• [6] AICTE PMSSS Portal — aicte-india.org/bureaus/jk

• [7] National Scholarship Portal (NSP) — scholarships.gov.in

• [8] JKBOPEE Official Website — jkbopee.gov.in

• [9] UGC / INSPIRE / DST Scheme Guidelines — ugc.gov.in, online-inspire.gov.in

• [10] AISHE Report 2023-24, Ministry of Education, Govt. of India

• Technology References:

• [11] LangChain Framework — docs.langchain.com

• [12] ChromaDB Vector Database — docs.trychroma.com

• [13] Streamlit Framework — docs.streamlit.io

• [14] Google Gemini API — ai.google.dev

• [15] all-MiniLM-L6-v2, Hugging Face — huggingface.co/sentence-transformers

This slide is NOT presented verbally. It sits at the end as a reference appendix. Judges may glance at it during Q&A.;

7. The 5-Minute Presentation Script

Word-for-word script calibrated to 4 minutes 50 seconds. Time each section with a stopwatch.

0:00 – 0:45 | SLIDE 1: PROBLEM

"Good morning, judges. Jammu & Kashmir has over two million students in higher education — and fewer than 200 career counselors to serve all of them. That's one counselor for every ten thousand students. In remote districts like Kupwara, Kargil, and Doda, students navigate scholarship deadlines, college cutoffs, and career decisions entirely alone. Their phones run on 2G. Generic AI chatbots hallucinate J&K-specific; data — they invent cutoff marks, fabricate deadlines. And 15% of students drop out because no one flagged the warning signs. What if we could give every one of these students an AI career counselor that works on 2G, uses only verified government data, and costs nothing?"

0:45 – 1:30 | SLIDE 2: SOLUTION

"We built J&K; EduSetu — 'Setu' means bridge in Hindi and Urdu. It's an AI-powered career intelligence platform built specifically for J&K; students. Three things make it different from anything that exists: First — it works on 2G networks. Responses come in under ten milliseconds, offline, with zero API calls. Second — every answer is grounded in verified government documents. No hallucinations. Every claim is cited with source and page number. Third — it's completely free. A student needs nothing more than a basic smartphone browser. It has seven integrated modules: AI advisor, college explorer, scholarship matcher, job navigator, mock interview simulator, resume analyzer, and an institutional admin portal. Let me show you."

1:30 – 3:00 | SLIDE 3: LIVE DEMO

"[Switch to localhost app] I'm in 2G mode. I'll type 'PMSSS scholarship'. [Type] — Answer appears in 0.2 milliseconds. Zero API calls. Official AICTE portal link included. Now the Scholarship Engine. I'll fill a profile: PCM stream, ■1.8 lakh income, Open Merit, female, 89%, age 18. [Click Match] — Five scholarships. PMSSS, INSPIRE at ■80K per year, Central Sector, Post-Matric, AICTE Pragati for girls. Colleges tab — here's the interactive J&K; map with 26 institutions pinned. NIT Srinagar, all the GMCs, IUST. This seat distribution bar is color-coded: blue for Open Merit, orange for SC, purple for ST. Language toggle — switching to Urdu. [Toggle] — entire interface updates. Mock interview — JKSSB Junior Assistant. [Submit one answer] — AI scores: Accuracy 72, Clarity 85, Completeness 68, Confidence 79."

3:00 – 3:45 | SLIDE 4: ARCHITECTURE

"Under the hood: a dual-path execution engine. Path one is a 2G offline cache — 22 pre-computed government topics, sub-ten-millisecond keyword matching. Path two is a full RAG pipeline — ChromaDB retrieves document chunks, Gemini generates cited answers. Three innovations: One — MD5 manifest fingerprinting reduced startup by

99.87%, from 28 seconds to 35 milliseconds. Two — a three-tier model router classifies query complexity and routes to the cheapest Gemini tier. Three — a four-layer fallback: Gemini to Groq to ChromaDB direct to offline cache. The student never sees a crash. Everything is tested — 19 automated tests, all passing."

3:45 – 4:30 | SLIDE 5: IMPACT (AISHA)

"Let me show you what this means for a real student. Aisha Bano, 18 years old, Kupwara district. She scored 89% in her boards. Before EduSetu: no counselor, 2G timeouts, ChatGPT gave her a wrong deadline, she had no idea she qualified for any scholarships, and her father told her to drop out. After EduSetu: she discovers five scholarships worth six point two lakh rupees in under ten minutes on a 2G phone. She explores NIT Srinagar's seat matrix, practices a mock interview scored by AI, gets her resume reviewed. Her father sees the PMSSS confirmation and supports her decision. Fifteen scholarships matchable. Twenty-two colleges mapped. Thirty careers profiled. Three languages. Sub-ten-millisecond offline. Zero cost."

4:30 – 5:00 | SLIDE 6: CLOSE

"EduSetu's architecture is state-agnostic. Replace the J&K; knowledge base with Rajasthan, Chhattisgarh, or Northeast India — the engine generalizes. Two million students need a bridge. We built it. Thank you."

8. Live Demo Strategy

Pre-Demo Checklist

- Run the app locally: **streamlit run app.py** on your laptop
- Open browser to **http://localhost:8501** and pre-load the app before your slot
- Pre-fill Aisha's profile in the Scholarship tab (but don't click Match yet)
- Set the mode to ■ **2G Ultra-Lite** for the first demo step
- Have a backup: Record a 90-second screen recording of the full demo flow
- Test the projector/HDMI 30 minutes before your slot. Set browser zoom to **90%**

Demo Flow (6 Steps in 90 Seconds)

Time	Feature	What to Show & Say
0:00–0:15	**2G Instant**	Type 'PMSSS scholarship' → 0.2ms answer → point out zero-API badge and portal link
0:15–0:35	**Scholarship Engine**	■ tab → Fill: PCM, ■1.8L, OM, Female, 89%, 18 → 5 matches appear instantly
0:35–0:45	**College Map**	■ tab → Interactive J&K; map loads → 26 pins → Zoom to Srinagar cluster
0:45–0:55	**Seat Distribution**	Scroll to NIT Srinagar → color-coded OM/SC/ST/RBA bar → 'Transparent data'
0:55–1:05	**Trilingual**	Switch sidebar to ■■■■ → prompts and badges update → Switch back to English
1:05–1:30	**Mock Interview**	■ tab → JKSSB → Start → Submit pre-typed answer → AI rubric score appears

9. Anticipated Judge Questions & Model Answers

SIH judges typically ask 5–8 questions in a 3-minute Q&A.; Keep answers **under 30 seconds each**. Here are 12 likely questions:

Q1: "How is this different from ChatGPT?"

ChatGPT is a general-purpose LLM that hallucinates J&K-specific; data. EduSetu is a domain-specific RAG system grounded in 17 verified government documents. Every answer has a cited source, page number, and similarity score. Plus, we work offline on 2G — ChatGPT cannot.

Q2: "How does it work without internet?"

Dual-path architecture. Path 1: pre-computed cache of 22 high-demand topics, matched via token-level Jaccard scoring in <10ms with zero API calls. Path 2 (online): full RAG with ChromaDB + Gemini. The app auto-detects the best path.

Q3: "What if government schemes change?"

Document-based architecture. Replace the PDF in /docs → MD5 manifest detects the change → auto re-indexes only modified files on restart. Takes <30 seconds. For the 2G cache, it's a one-line dictionary update.

Q4: "Is this scalable to other states?"

Absolutely. The architecture is state-agnostic. Replace the /docs folder, update college_data.py and scholarship_engine.py with state-specific data — the RAG engine, UI, model routing, and offline architecture carry over unchanged.

Q5: "How do you handle student privacy?"

We store zero personal data. The scholarship profile exists only in browser session state and is cleared when the tab closes. No login, no registration, no PII. Admin portal uses demo-generated data for the hackathon.

Q6: "What's the dropout risk prediction based on?"

8 weighted factors: attendance, failing subjects, financial stress, remote district, first-gen status, scholarship coverage, engagement drop, and missed deadlines. Composite score maps to 4 tiers (Low/Medium/High/Critical) with automated interventions.

Q7: "How accurate is the mock interview scoring?"

4-dimension rubric: Accuracy, Clarity, Completeness, Confidence — each scored 0–25 by Gemini. Total out of 100. Adaptive difficulty: students scoring >80 get harder questions.

Q8: "What's your tech stack?"

Streamlit frontend, Google Gemini (3 tiers) with Groq fallback, ChromaDB with all-MiniLM-L6-v2 embeddings, pypdf, tiktoken, LangChain. 19 automated tests via pytest.

Q9: "How do you prevent API rate limit crashes?"

Three layers: (1) Smart router sends simple queries to cheapest Gemini tier, (2) follow-up detector serves from conversation context without API calls, (3) four-layer fallback: Gemini → Groq → ChromaDB direct → 2G cache. Student never sees a crash.

Q10: "What's the business model?"

Student-facing features are permanently free. Production model is B2G: J&K; Education Department licenses the admin portal (dropout risk analytics). Colleges and district education offices are the paying customers.

Q11: "Why not just build a static website?"

A static website can't do: conversational AI advising, dynamic 7-field scholarship matching, AI-scored mock interviews, PDF resume analysis, or predictive dropout risk scoring. EduSetu is an intelligent platform, not a directory.

Q12: "Who built this?"

Team Error404, NIE Mysuru, CSE Batch 2027. 11 Python modules, 1500+ lines of UI, 17 government documents indexed, 19 automated tests — built specifically for SIH25094.

10. Presentation Dos & Don'ts

■ DO

- **Open with the 2M vs 200 stat.** Let the disparity hook judges in 5 seconds.
- **Demo on localhost.** Cloud deployments can fail. Local is bulletproof.
- **Use hard numbers.** '5 scholarships worth ■6.2L' beats 'many scholarships'.
- **Keep Aisha to ONE slide only (Slide 5).** Don't stretch the story across 6 slides.
- **Make eye contact with judges.** Look at them, not the screen.
- **Practice the demo 10+ times.** Every click should be muscle memory.
- **Have a backup demo video.** Record a 90s walkthrough before the event.
- **Keep slides minimal.** Max 5–6 bullets per slide, large fonts, one visual each.
- **Answer Q&A; concisely.** 30 seconds max per answer. Don't ramble.

■ DON'T

- **Don't read from slides.** The slides are your visual aid, not your script.
- **Don't open with 'We are Team Error404'.** Credentials come on Slide 2.
- **Don't explain tech stack first.** Problem → Solution → Demo → Architecture.
- **Don't apologize.** Never say 'we couldn't complete' or 'this feature is buggy'.
- **Don't demo features that might fail.** Only demo what's rehearsed flawlessly.
- **Don't exceed 5 minutes.** Judges penalize overtime. Use a stopwatch.
- **Don't argue with judge feedback.** Say 'Great point, we'll incorporate that.'
- **Don't build the whole PPT around Aisha.** She is one compelling data point on one slide. The rest of your presentation should be feature-driven and technically rigorous.

11. Impact & Scalability Metrics

Metric	Value	Why It Matters
2G Offline Latency	**< 10ms**	Works in Kupwara, Kargil, Doda — districts with 2G-only coverage
Scholarship Schemes	**15**	Central + State: PMSSS, NSP, INSPIRE, MCM, Pragati, Saksham, etc.
Max Aid Matchable	**■6.2 lakh+**	For a female PCM student with <■2.5L income
Colleges Mapped	**26 (with coordinates)**	Across 10 districts with branch-wise seat matrices
Job Profiles	**30+**	JKSSB, JKPSC, JK Bank, Central, Private, Entrepreneurship
Knowledge Chunks	**494**	From 17 official government documents indexed in ChromaDB
Mock Interview Templates	**5**	JKSSB, KAS, Engineering, Medical Viva, Bank PO
Dropout Risk Factors	**8**	Attendance, grades, income, remote district, first-gen, engagement, etc.

Metric	Value	Why It Matters
Languages	**3**	English, Hindi (■■■■■■), Urdu (■■■■)
Automated Tests	**19/19 passing**	Error handling, offline engine, RAG pipeline
Startup Speedup	**99.87%**	28 seconds → 35 milliseconds via MD5 manifest fingerprinting
Student Cost	**■0**	No registration, no login, no payment

Scalability Roadmap

- **Phase 1 (Current):** J&K-focused;: 15 scholarships, 22+ colleges, 30+ jobs. Deployed on Streamlit Cloud.
- **Phase 2 (6 months):** Expand to Ladakh, Rajasthan, Chhattisgarh. Integrate NCS and NSP APIs for real-time scheme updates.
- **Phase 3 (12 months):** B2G licensing for state education departments. Real institutional data integration. Voice input for accessibility.

12. Research & References (IEEE Format — Copy to Slide 7)

Below is the complete reference list in IEEE citation format. Copy these directly to your PPT Slide 7. Organize in two columns: Academic Papers (left) and Government/Technology Sources (right).

Academic & Research Papers

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL-HLT*, pp. 4171–4186, 2019.

[3] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proc. EMNLP-IJCNLP*, pp. 3982–3992, 2019.

[4] T. Brown, B. Mann, N. Ryder, et al., "Language Models are Few-Shot Learners," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 1877–1901, 2020.

[5] Gemini Team, Google DeepMind, "Gemini: A Family of Highly Capable Multimodal Models," *arXiv preprint* arXiv:2312.11805, 2023.

[6] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

[7] A. Vaswani, N. Shazeer, N. Parmar, et al., "Attention Is All You Need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

Government & Institutional Data Sources

[8] All India Council for Technical Education (AICTE), "Prime Minister's Special Scholarship Scheme (PMSSS) for J&K; and Ladakh Students," aicte-india.org/bureaus/jk, accessed 2026.

[9] Ministry of Electronics & IT, Govt. of India, "National Scholarship Portal (NSP)," scholarships.gov.in, accessed 2026.

[10] J&K; Board of Professional Entrance Examinations (BOPEE), "JKCET Admission Data & Cutoff Records," jkbopee.gov.in, accessed 2026.

[11] Department of Science & Technology (DST), Govt. of India, "Innovation in Science Pursuit for Inspired Research (INSPIRE)," online-inspire.gov.in, accessed 2026.

[12] University Grants Commission (UGC), "UGC NET JRF Scheme Guidelines," ugc.gov.in, accessed 2026.

[13] Ministry of Education, Govt. of India, "All India Survey on Higher Education (AISHE) Report 2023-24," aishe.gov.in, 2024.

[14] Ministry of Minority Affairs, Govt. of India, "Maulana Azad National Fellowship (MANF) & Merit-Cum-Means (MCM)," minorityaffairs.gov.in, accessed 2026.

Technology & Framework Documentation

[15] LangChain, "LangChain: Building Applications with LLMs through Composability," docs.langchain.com, accessed 2026.

[16] Chroma, "ChromaDB: The AI-native Open-Source Embedding Database," docs.trychroma.com, accessed 2026.

[17] Streamlit Inc., "Streamlit: The Fastest Way to Build Data Apps," docs.streamlit.io, accessed 2026.

[18] Google, "Gemini API Developer Documentation," ai.google.dev, accessed 2026.

[19] Groq Inc., "Groq Cloud API Documentation," console.groq.com/docs, accessed 2026.

[20] Hugging Face, "all-MiniLM-L6-v2: Sentence Transformer Model," huggingface.co/sentence-transformers/all-MiniLM-L6-v2, accessed 2026.

[21] OpenAI, "tiktoken: BPE Tokenizer for OpenAI Models," github.com/openai/tiktoken, accessed 2026.